

End-to-end scores for the cascade: definition and testing

Goal

Given an observed cloud x and the cascade’s fit $(\hat{f}, \hat{\theta})$, we ask whether the fitted model reproduces x . We simulate patterns from the fit and compare them with x using a proper scoring rule (lower is better). All models are scored on the same cloud, so only the fit differs.

Scores

Kernel score (primary). Let ϕ_i be the local configurations of x : the points within $R = 1.5$ mean spacings of a centre (one of x ’s own points), translated to the origin and divided by R . Let Φ, Φ' be configurations from *different* simulated patterns of model P . With a kernel k ,

$$S(P, x) = \frac{1}{2} \mathbb{E} k(\Phi, \Phi') - \frac{1}{m} \sum_{i=1}^m \mathbb{E} k(\phi_i, \Phi).$$

For a characteristic k this is strictly proper for the law of radius- R configurations: $\mathbb{E}_Q S(P, \cdot) = \frac{1}{2} \|\mu_P - \mu_Q\|^2 + \text{const.}$ Each configuration is embedded as a smoothed 21×21 grid ($h = 0.1$), so k is a Gaussian kernel of Euclidean distances between embeddings. Its width is set once per cloud, to the median distance between x ’s own configurations, and reused for every model, which is what makes scores comparable. We use $m \leq 300$ centres and 16 simulations per model.

Dawid–Sebastiani score (secondary). With T a vector of scale-free geometric statistics (Voronoi cell areas, Delaunay neighbour counts and angles; the persistent-homology statistics are dropped to avoid overlap with the PersLay models being scored), and μ_P, Σ_P the mean and shrunk covariance of T over 150 simulations,

$$S(P, x) = \log \det \Sigma_P + (T(x) - \mu_P)^\top \Sigma_P^{-1} (T(x) - \mu_P).$$

It is proper but not strictly (models agreeing on two moments tie).

Reference models and derived quantities. Two reference models are simulated per cloud: the *oracle* (true family, true θ) and *CSR* (Poisson with $\bar{n} = n$). Then

$$\text{regret} = S(\text{fit}) - S(\text{oracle}), \quad \text{gain} = S(\text{CSR}) - S(\text{oracle}), \quad \text{skill} = 1 - \frac{\sum \text{regret}}{\sum \text{gain}},$$

pooled over clouds, and reported only where the mean gain exceeds 2 standard errors. Skill 1 means as good as the truth; skill 0 means no better than CSR.

How the scores were tested

Power check. On 300 test clouds (15 per family and $\tilde{\delta}$ bin, plus 60 Poisson clouds), only the oracle and CSR were scored. A usable score should have gain > 0 where structure is strong, and gain ≈ 0 on Poisson. We report $d' = \overline{\text{gain}}/\text{sd}$ per cloud. The first score tried, a Wasserstein-1 energy score between whole patterns, had no power; the two scores above are its replacements.

d'	Thomas	nested	LGCP	Matérn II	Poisson (floor)
Kernel, $R=1.5$, point centres	1.19	1.20	1.30	1.06	0.12
DS, non-PH statistics	0.74	0.64	1.00	0.91	0.01
Wasserstein-1 energy (failed)	0.64	0.00	0.86	0.04	-0.13

Table 1: Power check, $\tilde{\delta} \geq 4$ bin (strongest structure); 15 clouds per family cell.

Findings. (i) At $\tilde{\delta} \geq 4$ the kernel score separates the truth from CSR for every family ($z = 4.1$ – 5.0). (ii) For $\tilde{\delta} < 1$ it is mostly noise, and nested at $\tilde{\delta} < 0$ goes the wrong way ($z = -2.1$); DS is no better there. (iii) Other kernel settings were biased on Poisson clouds (e.g. $R = 4$ with grid centres, $z \approx -2.6$); the chosen setting was picked after seeing these results, on the same clouds.

End-to-end run. Scoring the assembled cascade on 300 clouds gave regret near zero and skill between 0.94 and 1.22. Skill above 1 means the fit scored better than the truth, so this is noise: regret standard errors are as large as the regrets. The score cannot yet distinguish a good fit from the truth at this sample size.

Missing. There is no negative control, i.e. no test that a deliberately wrong fit (perturbed θ , wrong family) scores worse than the oracle. The power check only compares the truth with CSR, a large difference.